Python Collections

(Tuple)

Tuple

- A **tuple** is a collection which is:
  - **Ordered** (items have an index).
  - **Unchangeable (Immutable)** (cannot modify, add, or remove items).
- Tuples are written inside **round brackets ( )**.

Creating a Tuple

```
mytuple = ("apple", "banana", "cherry")
print(mytuple)  # ('apple', 'banana', 'cherry')
```

Accessing Items

```
mytuple = ("apple", "banana", "cherry")
print(mytuple[1])   # banana
print(mytuple[-1])  # cherry
```

Range of Indexes

```
mytuple = ("apple", "banana", "cherry", "orange", "kiwi", "melon")
print(mytuple[2:5])    # ('cherry', 'orange', 'kiwi')
print(mytuple[-4:-1])  # ('orange', 'kiwi', 'melon')
```

Count and Length

```
mytuple = ("apple", "banana", "apple", "cherry")
print(mytuple.count("apple"))  # 2
print(len(mytuple))          # 4
```

Loop Through a Tuple

```
mytuple = ("apple", "banana", "cherry")
for item in mytuple:
    print(item)
```

Check if Item Exists

```
mytuple = ("apple", "banana", "cherry")
if "banana" in mytuple:
    print("Yes, banana exists")
```

Workaround to Modify a Tuple

Since tuples are immutable, you can convert them to a list, modify, and convert back.

```
mytuple = ("apple", "banana", "cherry")
temp_list = list(mytuple)
temp_list[0] = "orange"
mytuple = tuple(temp_list)
print(mytuple)  # ('orange', 'banana', 'cherry')
```

Joining Tuples

```
tuple1 = ("a", "b", "c")
tuple2 = (1, 2, 3)
tuple3 = tuple1 + tuple2
print(tuple3)  # ('a', 'b', 'c', 1, 2, 3)
```

Key Point: Tuples are **immutable** → once created, they **cannot be changed**.

Summary

- **List:** Mutable (changeable), written with [ ].
- **Tuple:** Immutable (unchangeable), written with ( ).
- Use **lists** when you need to update items.
- Use **tuples** when data should stay constant.